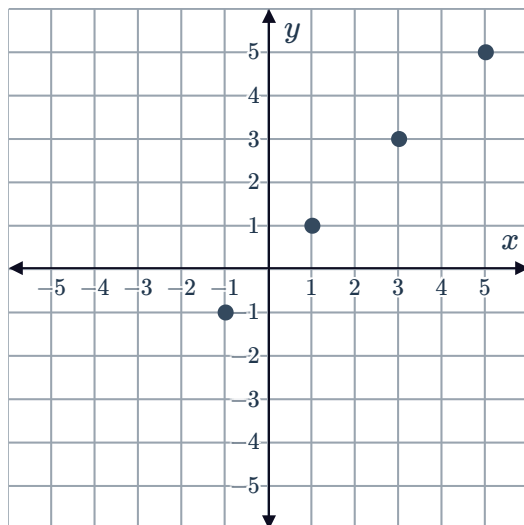


Linear relationships

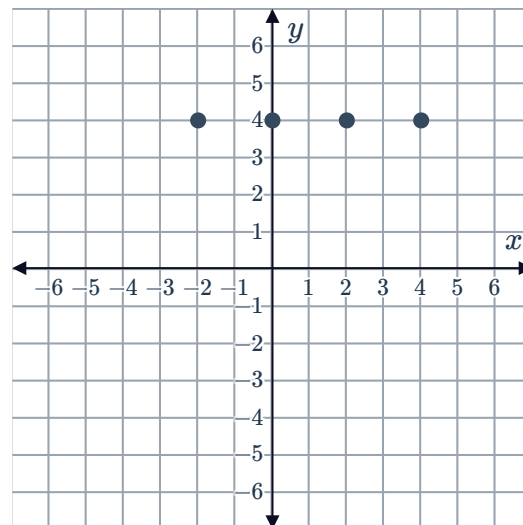


- 1 Do the points on each of the following number planes form a linear relationship? Explain your answer.

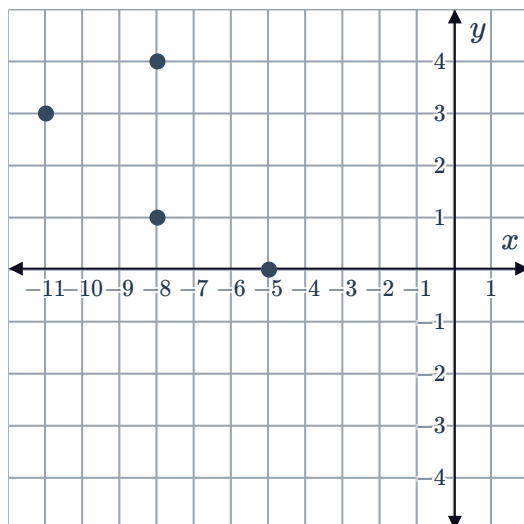
a



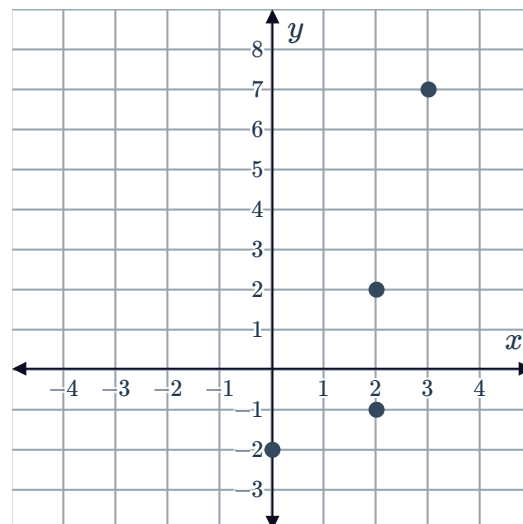
b



c



d



- 2 Consider the three points $A(3, 6)$, $B(1, 0)$, and $C(2, 3)$.

a Plot the points on a number plane.

b Do the points on each of the following number planes form a linear relationship? Explain your answer.

- 3 Consider the three points $X(3, -7)$, $Y(4, -9)$, and $Z(1, -2)$.

a Plot the points on the same number plane.

b Do the points on each of the following number planes form a linear relationship? Explain your answer.



Equations, tables and graphs

4 For each of the following equations and their table of values:

- i Plot the points from the table on a number plane.
- ii Determine whether the points form a linear relationship.

a $y = x + 5$

x	1	2	3	4
y	6	7	8	9

b $y = x$

x	1	2	3	4
y	1	2	3	4

c $y = 3$

x	-2	-1	0	1
y	3	3	3	3

d $y = x^2$

x	-2	-1	0	1
y	4	1	0	1

e $y = x + 4$

x	-4	-3	-2	-1
y	0	1	2	3

f $x = -6$

x	-6	-6	-6	-6
y	-6	-3	0	3

g $y = -x$

x	-3	-2	-1	0	1
y	3	2	1	0	-1

h $y = 4x$

x	-2	-1	0	1	2
y	-8	-4	0	4	8

5 For each of the following equations:

- i Complete the table of values provided.
- ii Plot the points from the table on a number plane.

a $y = 3x$

x	-1	0	1	2
y				

b $y = x - 5$

x	4	5	6	7
y				

c $y = x + 3$

x	1	3	7	10
y				



d $y = 2x + 7$

x	-2	-1	0	1
y				

e $y = 3$

x	-6	-3	-2	2
y				

f $y = -2x + 1$

x	-2	-1	0	1
y				

6 Consider the given table of values:

a Plot the points in the table on a number plane.

b Is the relationship between x and y linear?

x	-3	-2	-1	0
y	15	10	5	0

c If we extend the table of values to include $x = 1$, find the corresponding value of y .

7 Consider the given table of values:

a Plot the points in the table on a number plane.

b Is the relationship between x and y linear?

x	-8	-4	0	4
y	-2	-1	0	1

c If we extend the table of values to include $x = 8$, find the corresponding value of y .

8 Consider the given table of values:

a Plot the points in the table on a number plane.

b Is the relationship between x and y linear?

x	-1	0	1	2
y	-3	1	5	9

c If we extend the table of values to include $x = 3$, find the corresponding value of y .

9 Consider the given table of values:

Does the ordered pair $(-4, 8)$ satisfy the linear relationship between x and y ?

x	-2	-1	0	1
y	2	-1	-4	-7



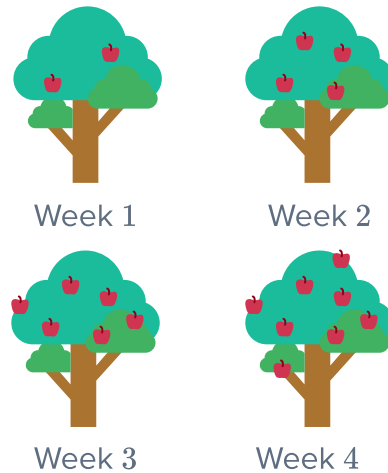
Applications

10 After each week, the number of apples on an apple tree increases according to the diagram:

a Complete the following table of values:

Week	1	2	3	4
No. Apples				

- b** Plot the number of apples against each week on a number plane.
- c** Is the relationship between the number of weeks and the number of apples linear?

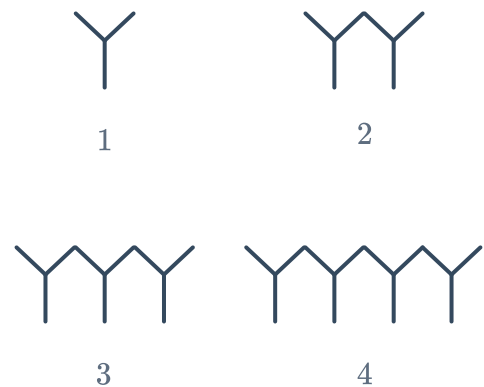


11 The following pattern was created using matchsticks. Each figure is labelled with a number:

a Complete the following table of values:

Figure number	1	2	3	4
No. Matchsticks				

- b** Plot the number of matchsticks used at each step on a number plane.
- c** Is the relationship between the step number and the number of matchsticks linear?

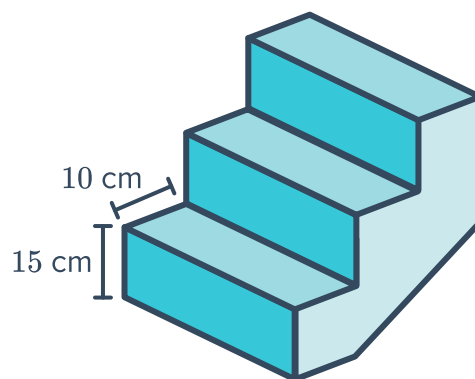


- 12** A staircase with 5 steps is shown below. The height of each step is 15 cm and the length of each step is 10 cm.



- a** Complete the following table of values:

No. Steps	2	3	4	5
Height of staircase (cm)	30			



- b** Plot the height of a staircase against the number of steps on a number plane.
- c** Is the relationship between the height of a staircase and the number of steps linear?
- d** Is it possible to have a staircase with height 105 cm with steps that have the same height and length as in the image?
- e** Complete the following table of values:

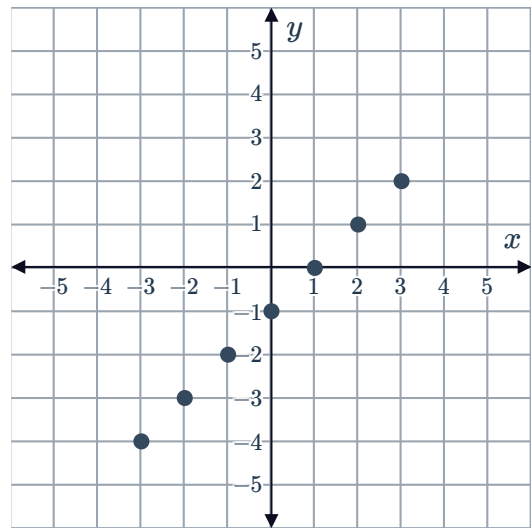
No. Steps	2	3	4	5
Length of staircase (cm)	20			

- f** Plot the length of a staircase against the number of steps on a number plane.
- g** Is the relationship between the length of a staircase and the number of steps in a staircase linear?
- h** Is it possible to have a staircase with length 70 cm with steps that have the same height and length as in the image?

- 13** A water tank initially starts with 65 L of water. Each day, 10 L of water is taken from the water tank.

- a** Plot the volume (in L) of water in the tank against the number days, from 0 to 4 days.
- b** Is the relationship between the volume of water in the water tank and the number of days linear?
- c** How many litres of water will there be after 5 days?

- 14 The points on the number plane show the relationship between water temperature, x , and surface air temperature, y :



- a Complete the table of values:

Water Temperature in $^{\circ}\text{C}$, x	-3	-2	-1	0	1	2	3
Surface Air Temperature in $^{\circ}\text{C}$, y							

- b Find the surface air temperature when the water temperature is 4°C .
c Find the surface air temperature when the water temperature is 14°C .

- 15 Let the height of a candle be h cm. If the candle is lit, the height of the candle decreases according to the equation $h = -2t + 9$, where t is the time in minutes. A table of values is shown:

Time (t mins)	0	1	2	3
Height of candle (h cm)	9	7	5	3

Plot the points in the table on a number plane.

- 16 A dead tree is 7 m tall and its height does not change over time. Let the height of the tree be y metres and the time passed be x years.

- a State whether the following coordinates satisfy the growth of the tree, where the x -coordinate is the time passed, and the y -coordinate is the height of the tree:

- i (3, 6) ii (0, 7) iii (7, 21) iv (10, 1)
v (4, 7) vi (2, 7) vii (7, 0) viii (7, 7)

- b Plot the points from part (a) that satisfy the growth of the tree on a number plane.

- 17 Let the volume of water in a tank be V L,  the time taken to fill up the tank be t minutes. Suppose the tank is initially empty and the number of litres in the tank is always three times the number of minutes that have passed.

a Complete the table of values:

Time (t mins)	0	1	2	3
Volume of water (V L)				

b Which of the following equations represents the relationship between the time taken, t , and the volume of the tank, V ?

A $V = 3$ **B** $V = 3t$ **C** $V = 3t + 2$ **D** $V = t + 3$

c Plot the points in the table on a number plane.

- 18 Let the height of a pot plant be y cm. As the pot plant grows, the height increases according to the equation $y = 2t + 9$, where t is the elapsed time in weeks.

a Complete the table of values:

Time (t weeks)	0	1	2	3
Height of pot plant (y cm)				

b Plot the points in the table on a number plane.

- 19 Max is 21 years old and Sebastian is 14 years old. Let M represent Max's age and S represent Sebastian's age.

a Write an equation for M in terms of S .

b State whether following ordered pairs, in the form (S, M) , satisfy the relationship between Max and Sebastian's ages:

- i (14, 21) ii (36, 29) iii (20, 27) iv (25, 25)
v (24, 31) vi (16, 23) vii (38, 46)

c Plot the points from part (b) that satisfy the relationship between their ages on a number plane.